

19-200-0105B BASIC MECHANICAL ENGINEERING

MODULE 1

- Thermodynamic systems – open, closed and isolated systems
- Equilibrium state of a system
- Thermodynamic property ,state, process, and cycle
- Thermodynamic process
- Thermodynamic Cycle
- Zeroth law of thermodynamics- concept of temperature, temperature scales.
- First law – internal energy, enthalpy, work and heat
- Different processes- isobaric, isochoric, isothermal ,adiabatic processes and polytropic processes
- Second law – Kelvin-plank and Claussius statements and their equivalence
- Carnot Cycle.
- Thermodynamic properties of Steam , Steam Generator
- Different types of boilers
- boiler mountings and accessories
- Formation of steam at constant pressure
- working of steam turbines
- compounding of turbines

- **Mechanical engineering** is a branch of engineering science that deals with the design and development of work producing and work absorbing machines.
- **Thermodynamics** deals with the interaction between heat and work and the working substance employed to obtain energy conversion.

Thermodynamics

- Thermodynamics can be defined as the *science of energy* and its effect on the physical properties of substances.
- The name thermodynamics was formed from the Greek words therme (heat) and dynamics (power).
- Thermodynamics is a branch of Physics that deals with the relationship among heat, work and properties of system which are in equilibrium with one another.
- Thermodynamics deals with the change of one form of energy to another form.

Working

Substance

- The conversion of heat to work or work to heat takes place through agency called working substance.
- Any pure substance can be employed as a working substance.
- Air and steam are two main working substance usually employed in engineering practice.
- The working substance is capable of changing its volume so that it can be a medium for work transfer.

**Classical
Thermodynamics
(Macroscopic
approach)**

**Statistical
Thermodynamics
(Microscopic
approach)**

Thermodynamics

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graph TD; A([Thermodynamics]) --- B[Classical Thermodynamics (Macroscopic approach)]; A --- C[Statistical Thermodynamics (Microscopic approach)];
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Classical Thermodynamics & Statistical Thermodynamics

- Macroscopic approach to the study of thermodynamics that does not require a knowledge of the behaviour of individual particles is called **Classical Thermodynamics(Macroscopic approach)**. It provides a direct and easy way to the solution of engineering problems. Eg. Pressure, Temperature...Etc
- The matter is composed of molecules and the analysis is carried out by considering the position, velocity and energy of each molecules, is called **Statistical Thermodynamics(Microscopic approach)**. The effect of molecular motion is considered

Some application areas of thermodynamics



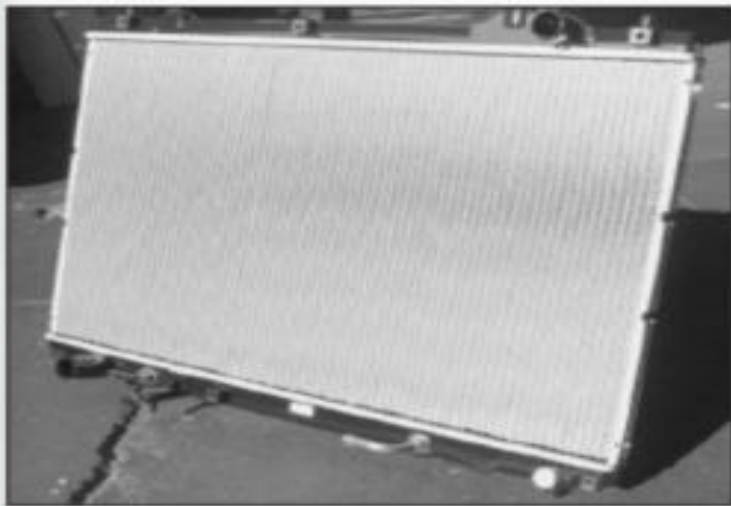
The human body



Air conditioning systems



Airplanes



Car radiators

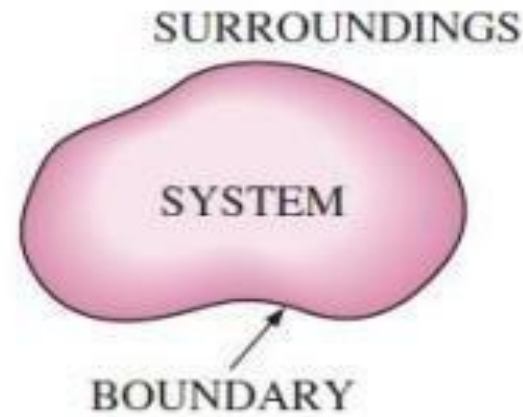


Power plants

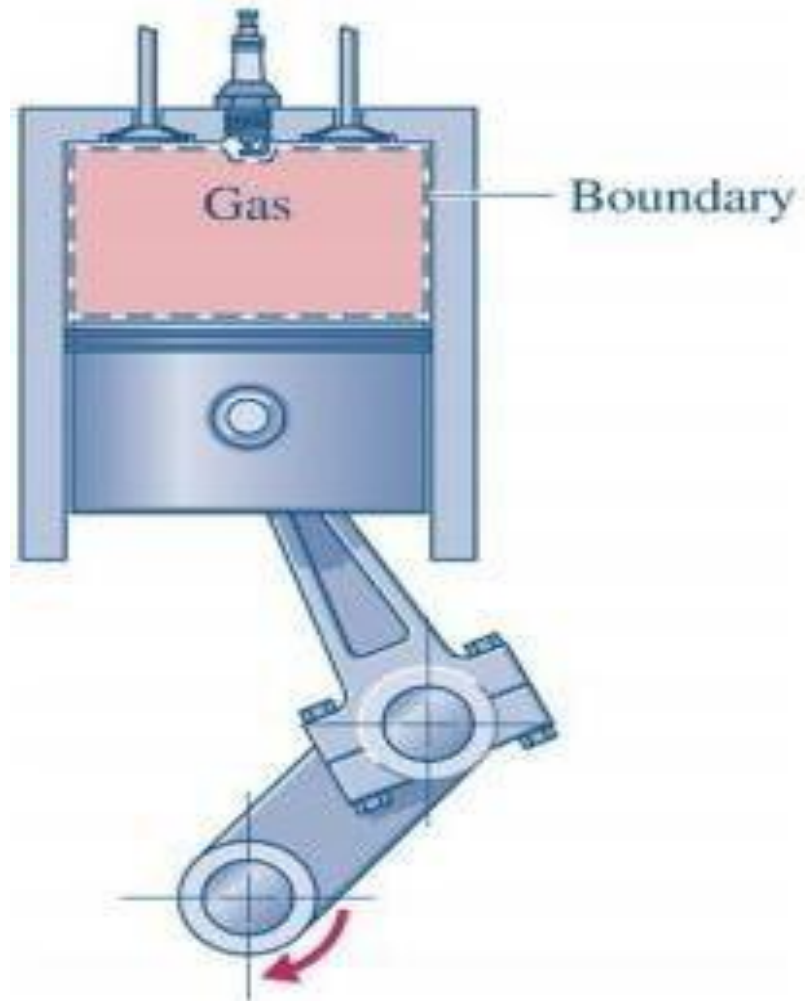


Refrigeration systems

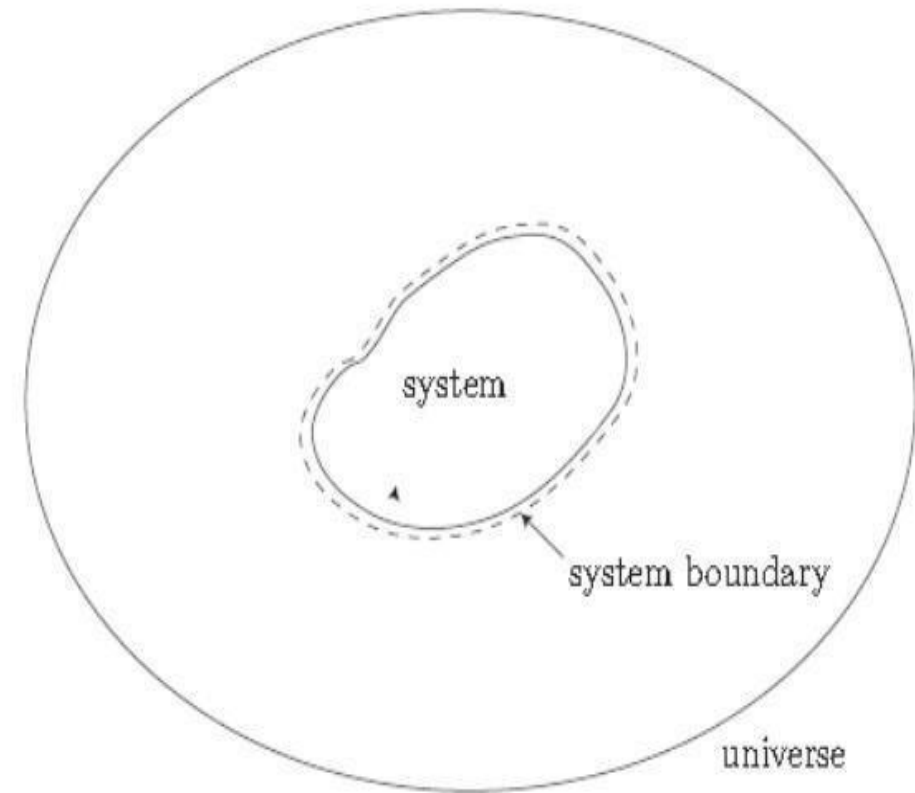
SYSTEMS



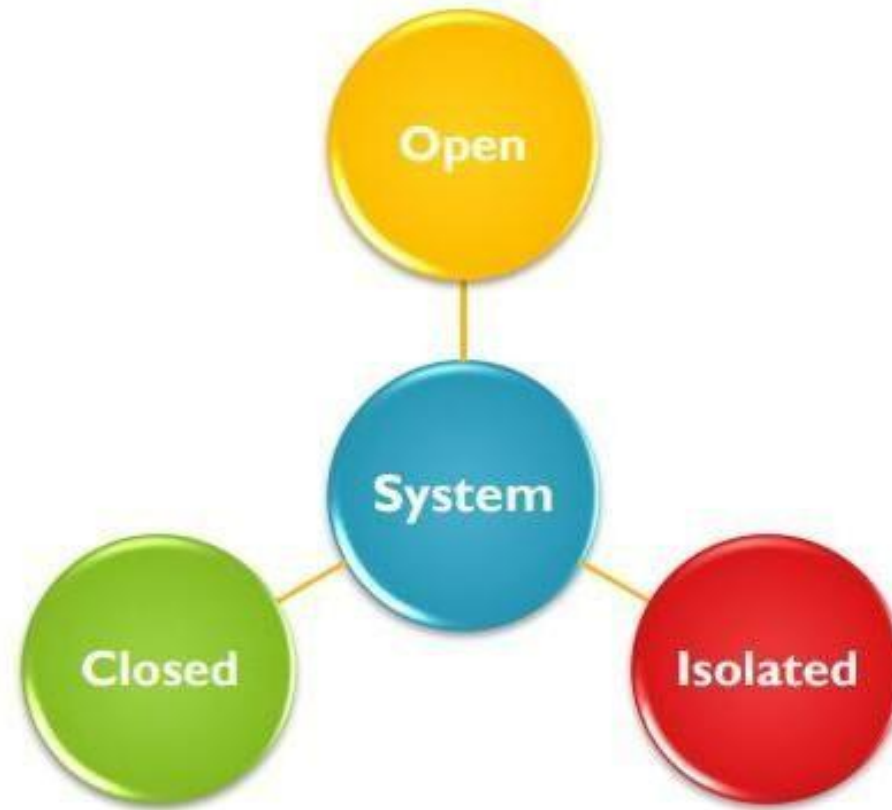
- A system is defined as a quantity of matter or a region in space chosen for study.
- The mass or region outside the system is called the **Surroundings**. The real or imaginary surface that separates the system from its surroundings is called the **Boundary**. The boundary of a system can be fixed or movable. Note that the boundary is the contact surface shared by both the system and the surroundings. Mathematically speaking, the boundary has zero thickness, and thus it can neither contain any mass nor occupy any volume in space.



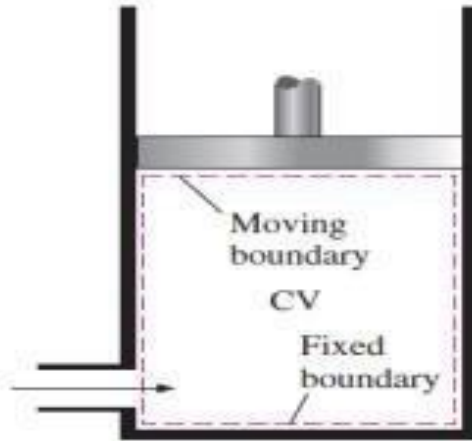
Universe: Combination of system and surroundings.



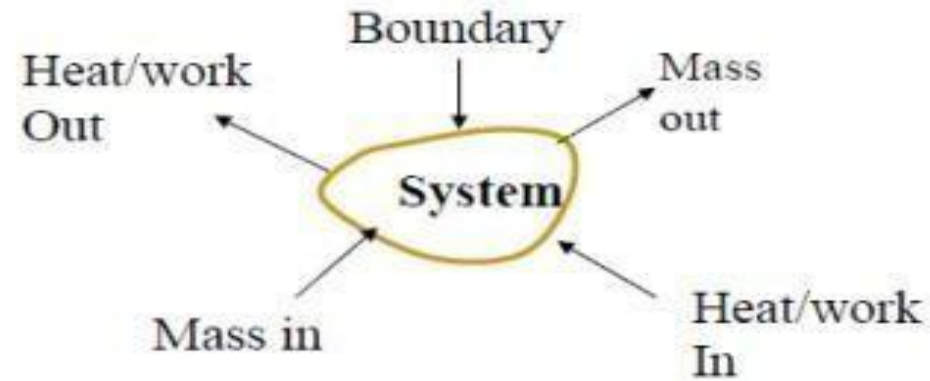
Types of systems



Open System(Control Volume)

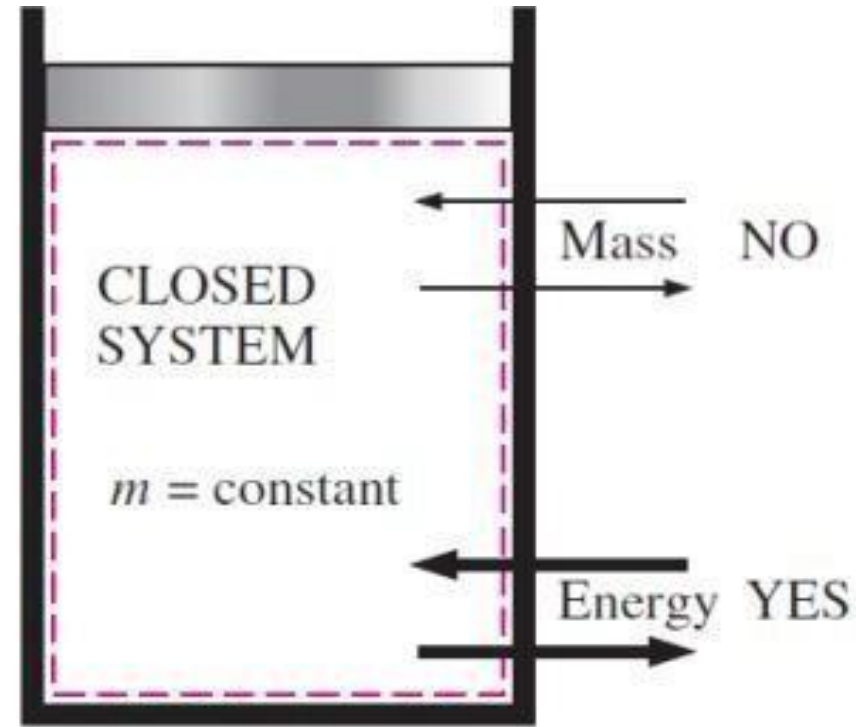
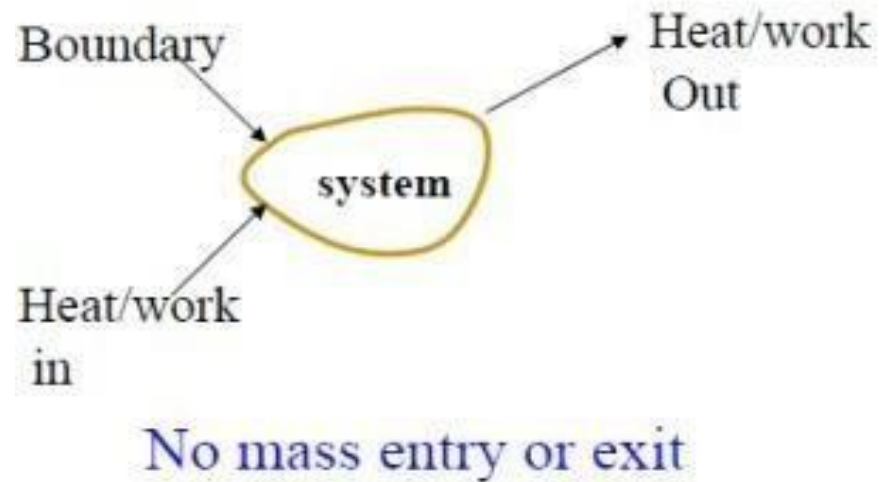


(b) A control volume with fixed and moving boundaries



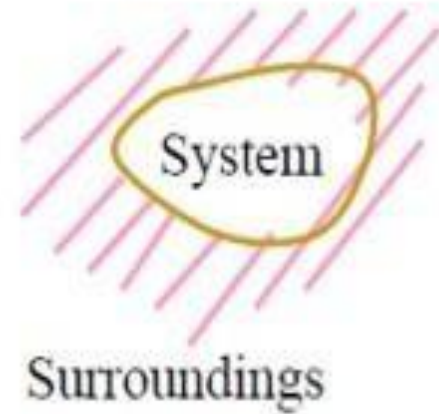
- Both mass and energy can cross the boundary of a control volume
- Eg. compressor, turbine, nozzle

Closed System



- Fixed amount of mass, and *no mass can cross its boundary*
- Energy in the form of heat or work, can cross the boundary; and the volume of a closed system does not have to be fixed

Isolated system



- There is no interaction between system and the surroundings. It is of fixed mass and energy, and hence there is no mass and energy transfer across the system boundary
- Eg. Universe

Equilibrium

- Thermodynamics deals with equilibrium states. The word equilibrium implies a state of balance. In an equilibrium state there are **no unbalanced potentials (or driving forces) within the system**. A system in equilibrium experiences no changes when it is isolated from its surroundings.
- A system is said to be in thermodynamic equilibrium , if it satisfies the following requirements of equilibrium

- ❖ Mechanical equilibrium
- ❖ Thermal equilibrium
- ❖ Chemical equilibrium
- ❖ Phase equilibrium

Equilibrium: A state of balance

- **Thermal equilibrium:** If the temperature is the same throughout the entire system.
- **Mechanical equilibrium:** If there is no change in pressure at any point of the system with time.
- **Phase equilibrium:** If a system involves two phases and when the mass of each phase reaches an equilibrium level and stays there.
- **Chemical equilibrium:** If the chemical composition of a system does not change with time, that is, no chemical reactions occur.

- **There is no unbalanced force acting on the system**

- **Temperature is same throughout the entire system**

**MECHANICAL
EQUILIBRIUM**

**THERMAL
EQUILIBRIUM**

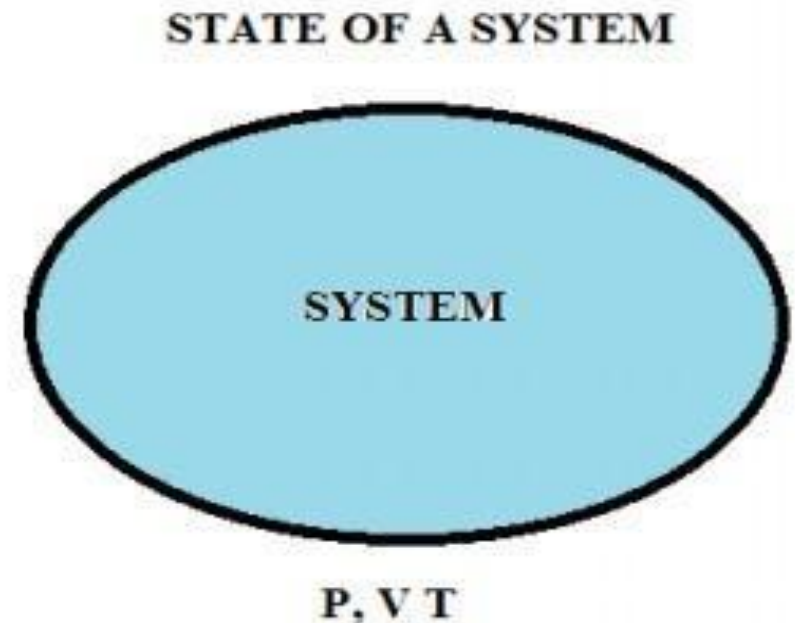
**PHASE
EQUILIBRIUM**

**CHEMICAL
EQUILIBRIUM**

- **Mass of each phase reaches an equilibrium level**

**Chemical
Composition
does not change
with time, that
is, no chemical
reactions occur**

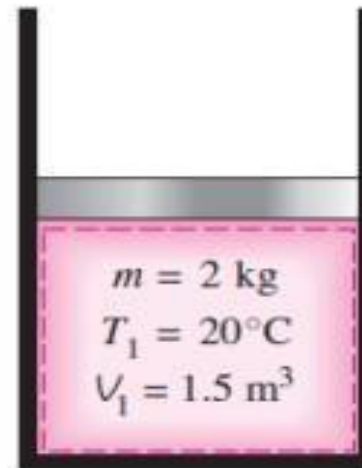
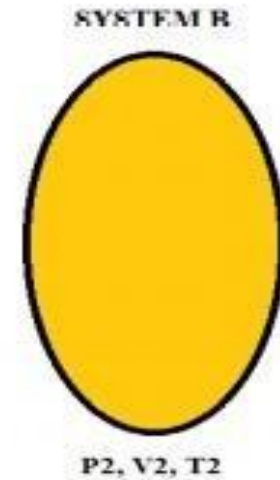
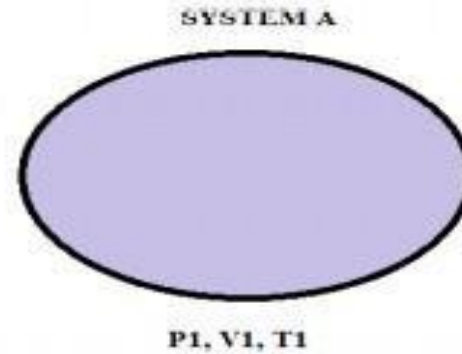
State of a System



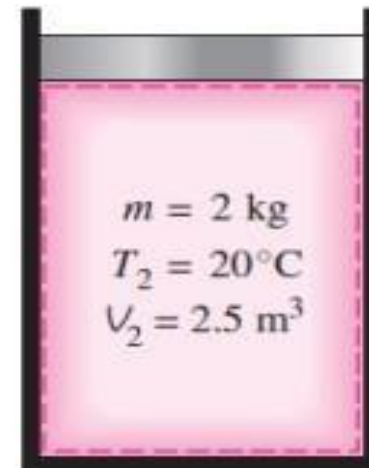
- Condition of physical existence of a system at any instant
- State of a thermodynamic system is described by specifying its thermodynamic co-ordinates or thermodynamic properties
- At a given state, all the properties of a system have fixed values

Properties of a System

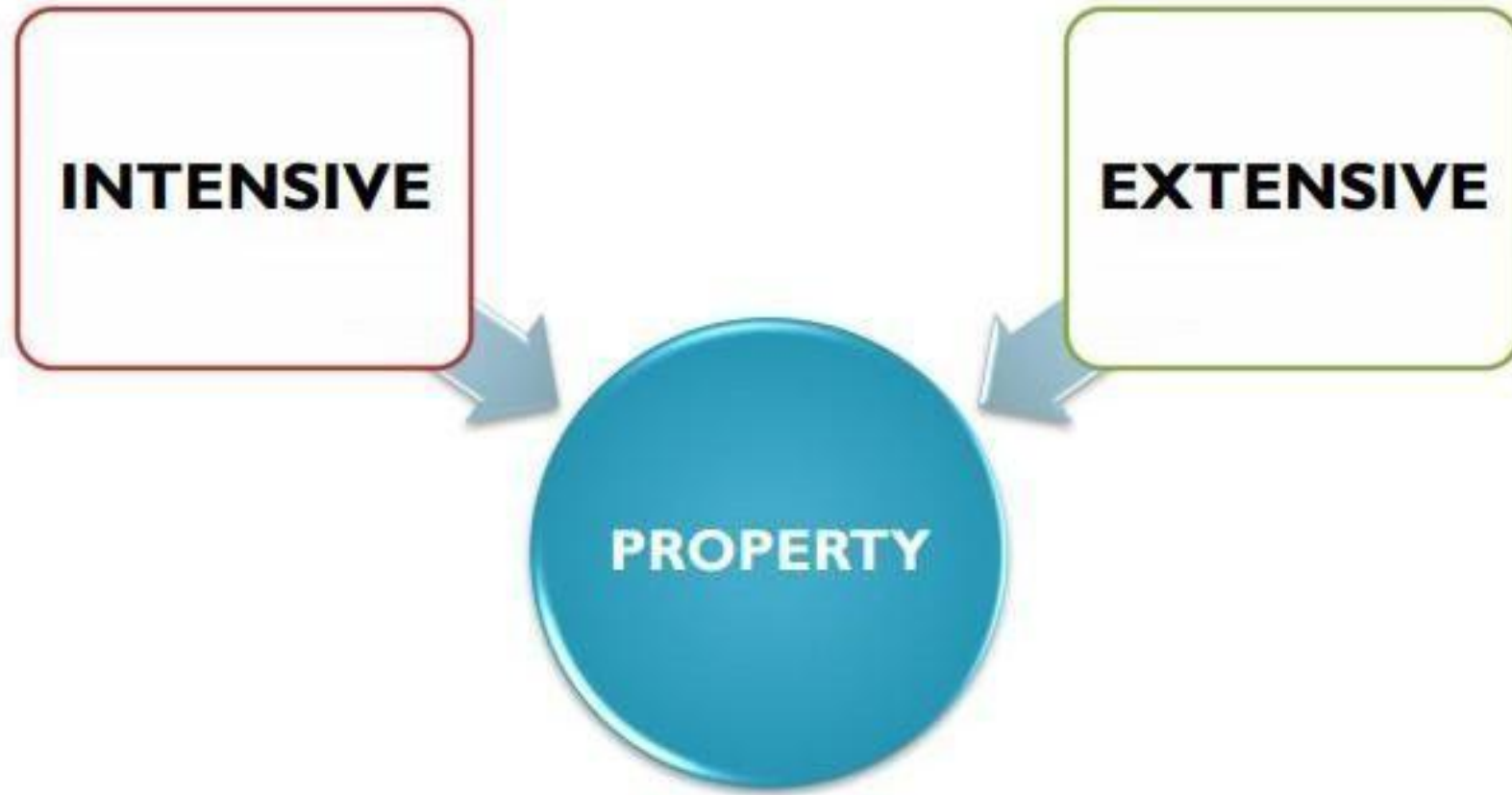
- Any characteristic of a system is called a property. Some familiar properties are Pressure P, Temperature T, Volume V, and Mass m.
- Quantities which identify the state of a system.
Property must have a definite value when the system is at a particular state.



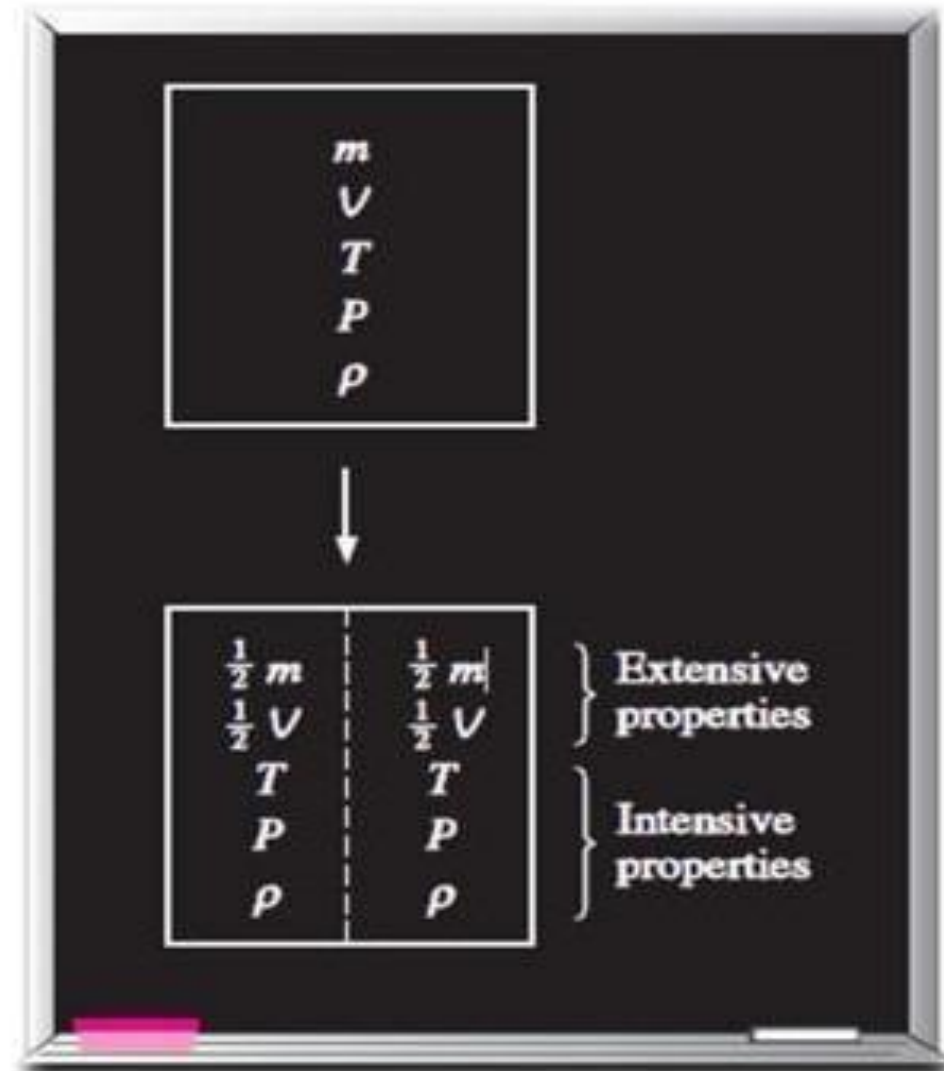
State 1



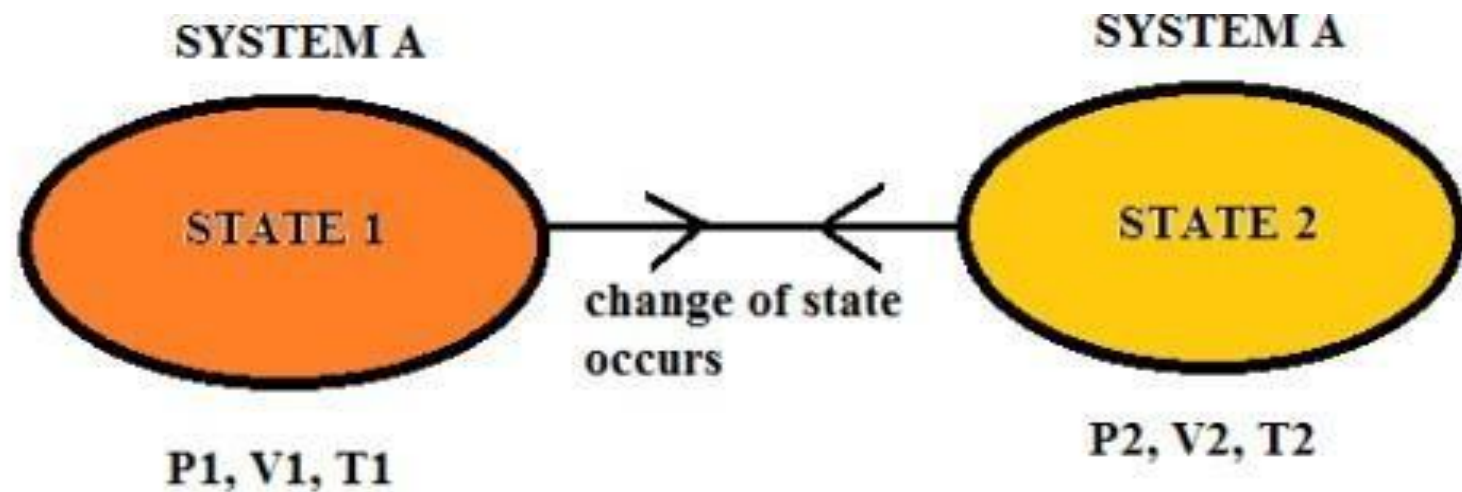
State 2



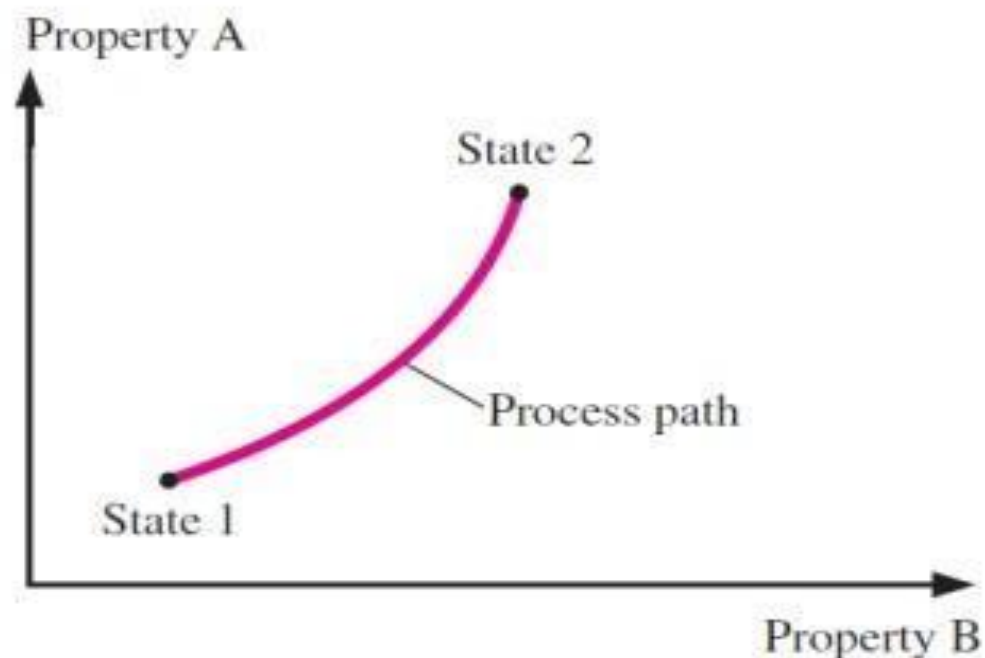
- Intensive properties are those that are **independent of the mass** of a system, such as temperature, pressure, and density
- Extensive properties are those whose values **depend on the size**—or extent—of the system. Total mass, total volume, and total momentum are some examples of extensive properties



PROCESS



- Any change that a system undergoes from one equilibrium state to another is called a **process**
- Transformation from one state to another

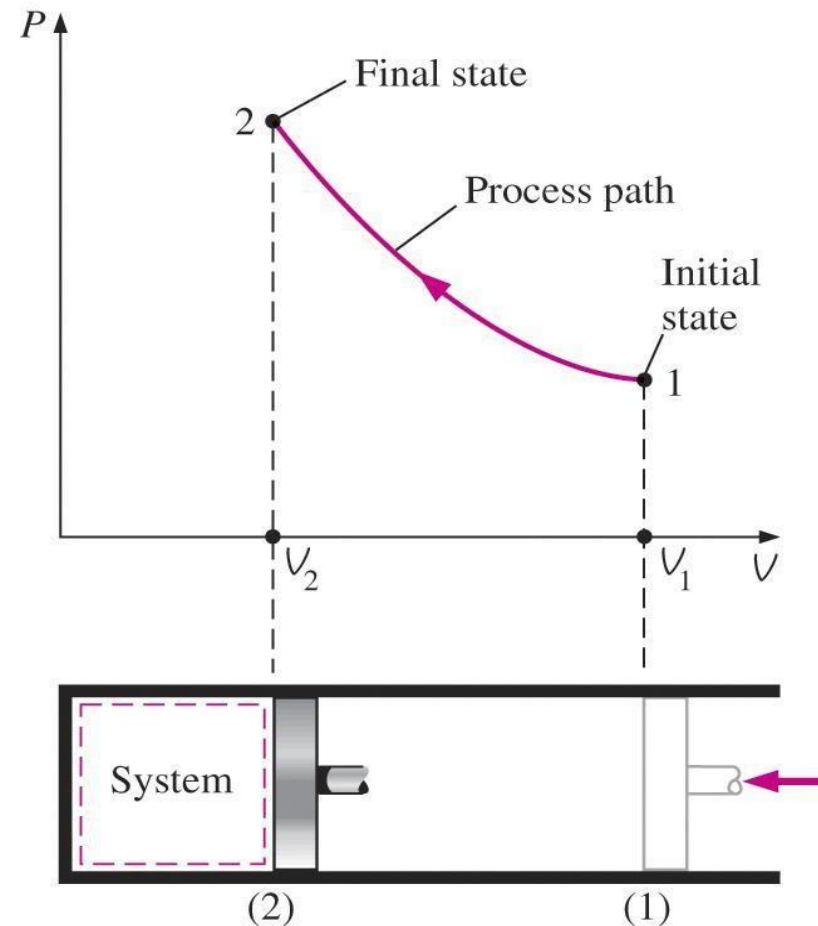


The transformation of a thermodynamic system from one state to another is called a **process**.

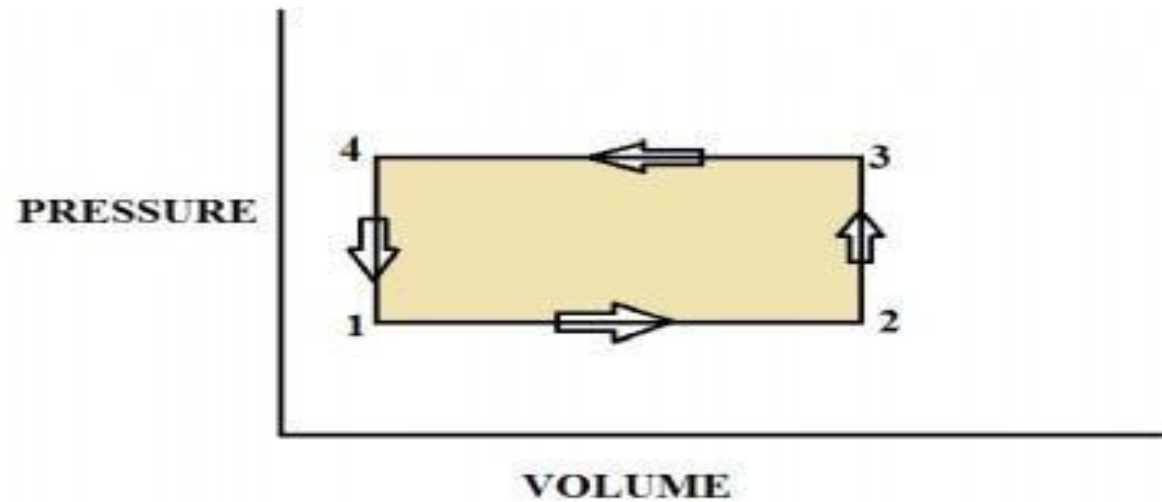
- Isothermal process
- Isobaric process
- Isochoric process
- Isenthalpic process
- Isentropic process
- Adiabatic process
- Reversible process
- Irreversible process
- Quasi -static process

PATH

- The series of states through which a system passes during a process.
- To describe a process completely, one should specify the initial and final states, as well as the path it follows, and the interactions with the surroundings.



CYCLE



- When a system in a given initial state experiences a series of processes and returns to the initial state, the system undergoes a cycle. At the end of the cycle the properties of the system have the same values they had at the beginning.
- Thermodynamic path in a cycle is in closed loop form
- **Cyclic integral of any property in a cycle is zero**
- $\oint dp = 0$; 'p' is any thermodynamic property

Path and Point Functions

- Path Functions

- ❖ If the value of the *thermodynamic variable depends upon the path* followed in going from one state to another
- ❖ Path functions are not properties of the system
- ❖ Path functions have inexact differentials designated by the symbol 'δ'
- ❖ Eg. Work(W), Heat(Q)

- ❖ $\int_1^2 \delta W = W_{12} \quad ; \text{ Not } (W_2 - W_1) \quad ; \text{ Not } dW$

- ❖ $\int_1^2 \delta Q = Q_{12} \quad ; \text{ Not } (Q_2 - Q_1) \quad ; \text{ Not } dQ$

- **POINT FUNCTIONS**

- ❖ Properties does not depend on the path followed in reaching the state, but only on the equilibrium state itself
- ❖ Point functions are properties of the system
- ❖ Point functions have exact differentials designated by the symbol 'd'
- ❖ Eg. Pressure, Volume, Temperature, Density, Enthalpy, Entropy

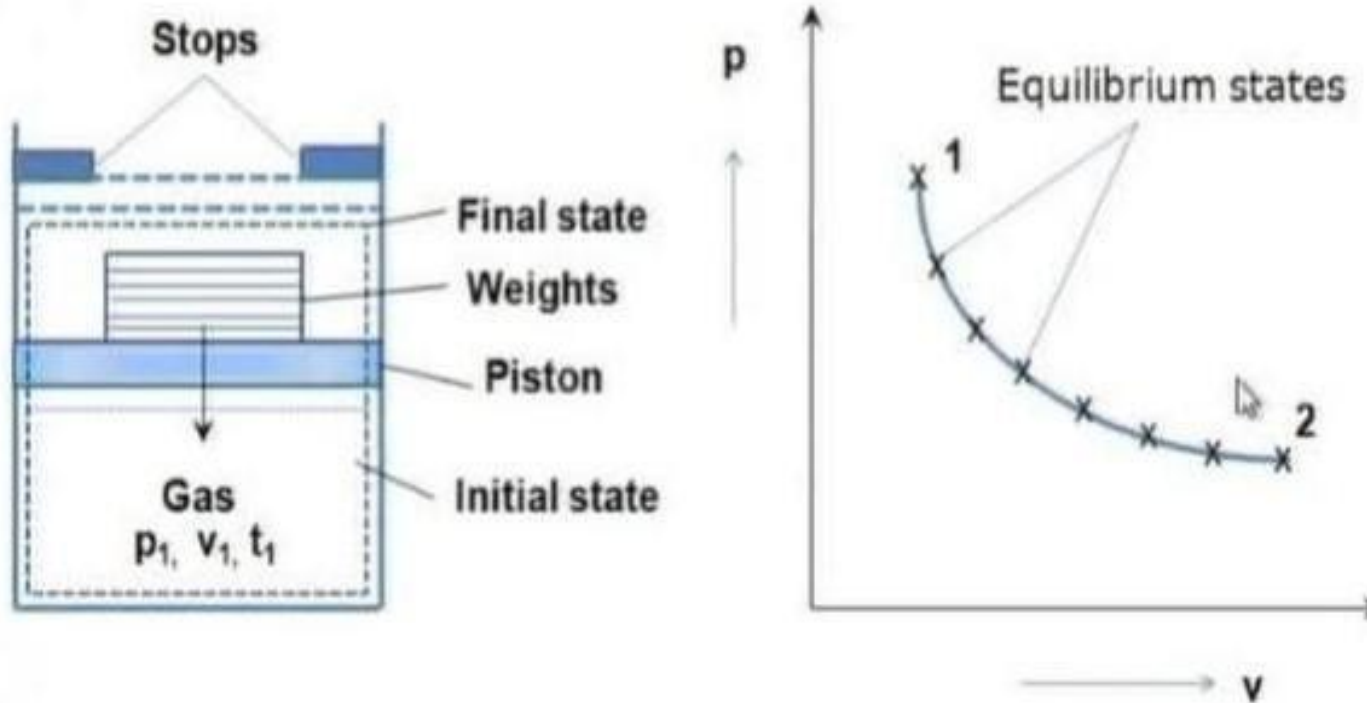
$$\int_1^2 dV = V_2 - V_1 = \Delta V$$

Quasi-static

Processes

The processes can be restrained or unrestrained, we need restrained processes in practice.

- A quasi-static process is one in which the deviation from thermodynamic equilibrium is infinitesimal. All states of the system passes through are equilibrium states.

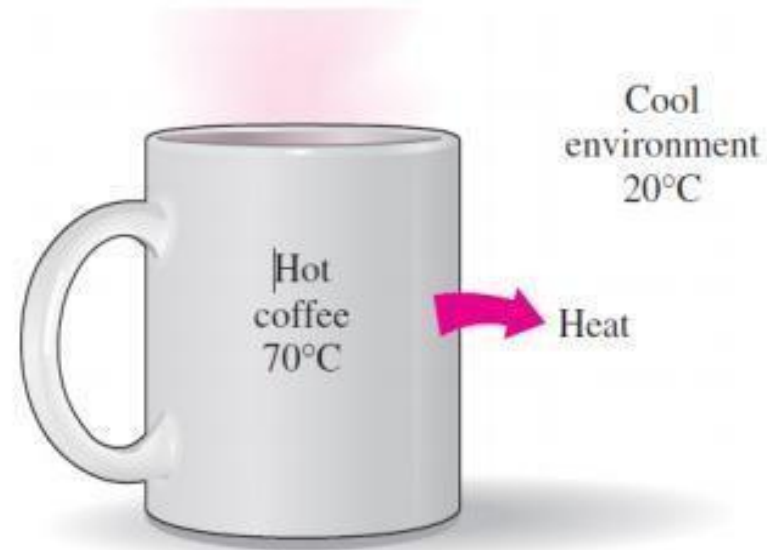


Quasi-static Processes

- If we remove the weights slowly one by one the pressure of the gas will displace the piston gradually. It is quasistatic.
- On the other hand if we remove all the weights at once the piston will be kicked up by the gas pressure. (This is unrestrained expansion) but we don't consider that the work is done - because it is not in a sustained manner. It is not quasistatic.
- In both cases the systems have undergone a change of state.

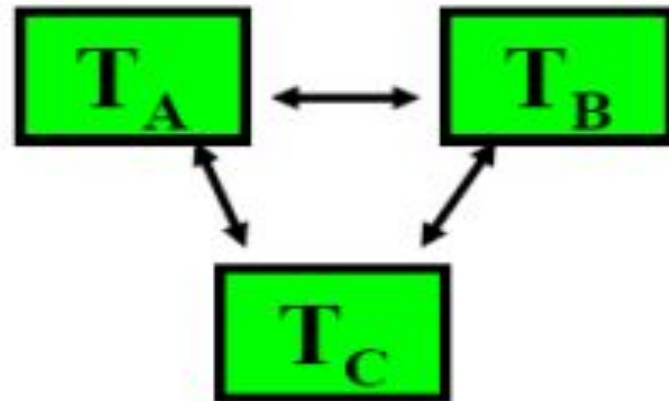
Temperature and The Zeroth Law of Thermodynamics

- Degree of hotness or coldness of a body or environment
- The equality of temperature is the only requirement for thermal equilibrium
- Driving potential causing the flow of energy as heat



Zeroth Law of Thermodynamics

- The zeroth law was first formulated and labelled by R. H. Fowler in 1931
- *If two systems (say A and B) are in thermal equilibrium with a third system (say C) separately (that is A and C are in thermal equilibrium; B and C are in thermal equilibrium) then they are in thermal equilibrium themselves (that is A and B will be in thermal equilibrium)*
- All temperature measurements are based on this LAW



SIGNIFICANCE OF ZEROth LAW OF THERMODYNAMICS

- The Zeroth law of thermodynamics is the basis for measurement of temperature and setting its scale.
- The zeroth law is incredibly important as it allows us to define the concept of a temperature scale.
- It is by the property of temperature that we can make out the difference between hot and cold body.
- The zeroth law of thermodynamics can be used to compare temperature of several bodies

Temperature Scales

- All temperature scales are based on some easily reproducible states such as the *freezing and boiling* points of water: the *ice point and the steam point*
- **Celsius scale**: in SI unit system
- **Fahrenheit scale**: in English unit system
- For engineers the more useful temperature scale is the absolute temperature scale.
- **Kelvin scale** (SI) **Rankine scale** (E)

Temperature Scales

- Temperature scales enable us to use a common basis for temperature measurements

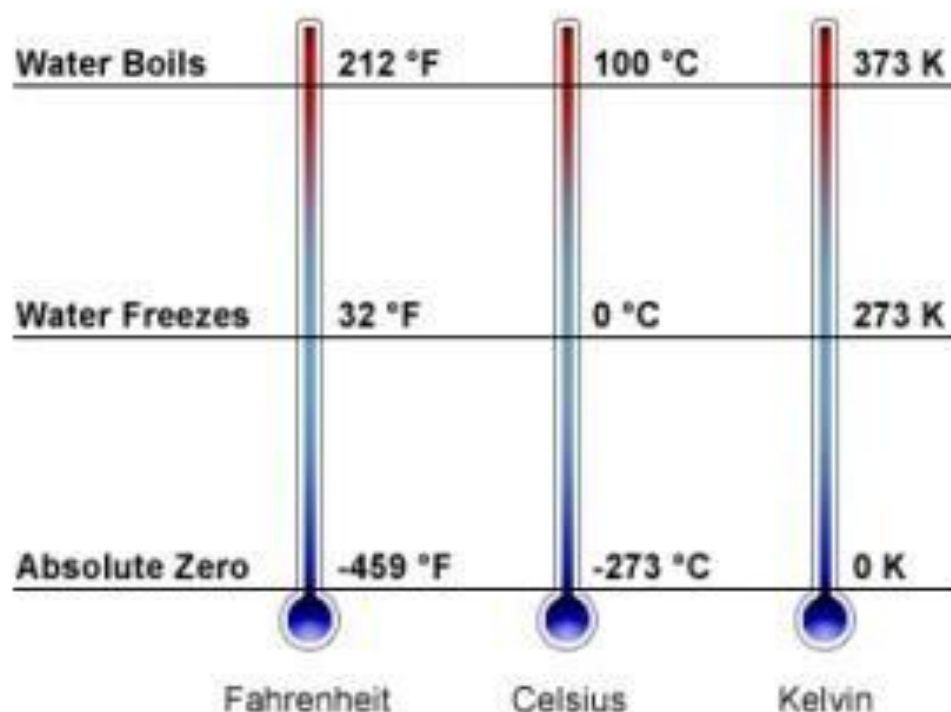
❖ **Celsius scale** (A. Celsius, 1702–1744); 0°C - 100°C

❖ **Fahrenheit scale** (G. Fahrenheit, 1686–1736); 32°F – 212°F ;
 $T(^{\circ}\text{F}) = 1.8 T(^{\circ}\text{C}) + 32$

❖ **Kelvin scale** (Lord Kelvin, 1824–1907); 273.15K - 373.15K ;
 $T(\text{K}) = T(^{\circ}\text{C}) + 273.15$

Absolute Zero

Thermometers compare Fahrenheit, Celsius, and Kelvin scales



Pressure

- Pressure is defined as a normal force exerted by a fluid per unit area

$$P = F/A$$

- The SI unit of pressure is the Pascal

$$(\text{Pa}) \quad 1 \text{ Pa} = 1 \text{ N/m}^2$$

named after Blaise Pascal (1623-1662).

- Another popular unit is “bar”

$$1 \text{ bar} = 10^5 \text{ N/m}^2$$

$$1 \text{ atm} = 1.01325 \times 10^5 \text{ N/m}^2$$

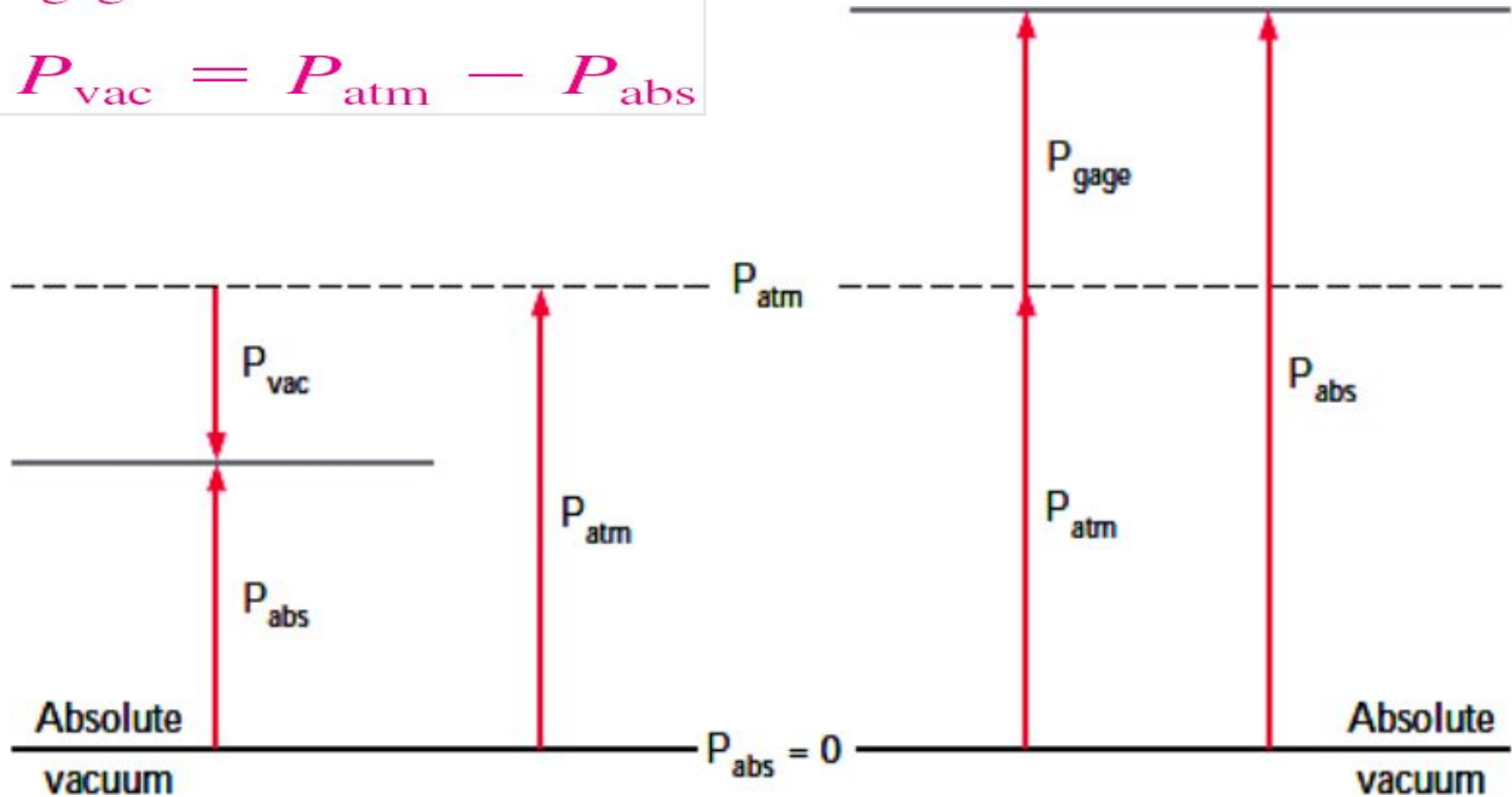
ABSOLUTE, GAUGE and VACUUM PRESSURES

- **Absolute pressure:** When the pressure is measured , with complete vacuum or absolute zero is taken as datum
- **Gage pressure:** local atmospheric pressure is taken as datum
The difference between the absolute pressure and the local atmospheric pressure. Most pressure-measuring devices are calibrated to read zero in the atmosphere, and so they indicate gage pressure.
- **Vacuum pressures:** pressure is measured below the atmospheric pressure

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}}$$
$$P_{\text{vac}} = P_{\text{atm}} - P_{\text{abs}}$$

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}}$$

$$P_{\text{vac}} = P_{\text{atm}} - P_{\text{abs}}$$



Forms of Energy



- Energy can exist in numerous forms such as thermal, mechanical, kinetic, potential, electric, magnetic, chemical, and nuclear, and their sum constitutes the **total energy E of a system**
- **Total energy, $E = \text{Macroscopic Energy} + \text{Microscopic Energy}$**
- Thermodynamics provides no information about the absolute value of the total energy. It deals only with the change of the total energy

- The macroscopic forms of energy are those a system possesses as a whole with respect to some outside reference frame, such as kinetic and potential energies
- The microscopic forms of energy are those related to the molecular structure of a system and the degree of the molecular activity, and they are independent of outside reference frames. The sum of all the microscopic forms of energy is called the **Internal energy** of a system and is denoted by **U**

$$E = U + \text{KE} + \text{PE} = U + m \frac{V^2}{2} + mgz \quad (\text{kJ})$$

- Energy change = Energy at final state - Energy at initial state

$$\Delta E_{\text{system}} = E_{\text{final}} - E_{\text{initial}} = E_2 - E_1$$